Hemang Varshney

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Summary

Aspiring **Quantitative Researcher** and **Software Engineer** with a strong foundation in C/C++, algorithms, probability, statistics, and machine learning. Passionate about mathematical problem solving, and analytical reasoning, with experience exploring competitive programming and quantitative concepts. Seeking internship opportunities in quantitative research, software engineering, or risk analytics.

Education

University of Delhi — Faculty of Technology **Aug 2025 – Aug 2029**

Bachelor of Technology in Computer Science and Engineering (8 SGPA) *Delhi, India*

- Relevant Coursework: Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Discrete Mathematics

Work Experience

Quantitative Risk Analyst Intern **Jan 2026 – Mar 2026**

Zetheta Algorithms Private Ltd *Remote*

- Engineered **3+ quantitative risk dashboards** in Python and R, reducing manual reporting time by **40%** through automated simulation pipelines.
- Designed and validated **4 probabilistic decision-theoretic and causal inference models**, improving model accuracy by identifying key structural assumptions.
- Conducted **scenario-based sensitivity analysis** across **8+ market conditions**, quantifying model robustness and exposure behaviour to inform risk-aware trading decisions.

Projects

Kelly Betting Simulation | *Python, NumPy, Matplotlib, Monte Carlo Methods*

- Built a **Monte Carlo simulation framework** executing **10,000+ iterations** to evaluate Kelly Criterion-based betting strategies, benchmarking **5 distinct allocation strategies** (full Kelly, fractional Kelly, fixed-stake variants).
- Quantified long-term wealth growth, maximum drawdown, and volatility metrics; demonstrated that a **half-Kelly strategy reduced drawdown by up to 35%** compared to full-Kelly under probability misestimation scenarios.

Probability Calibration Analysis | *Python, Scikit-learn, Pandas, Reliability Diagrams*

- Evaluated **3 calibration methods** (Platt Scaling, Isotonic Regression, Histogram Binning) across **1,000+ prediction samples**, systematically diagnosing overconfidence and underconfidence in probabilistic classifiers.
- Produced calibration curves and Brier score breakdowns that improved model interpretability for non-technical stakeholders.

Value of Information (VOI) Analysis | *Python, Decision Theory, Utility Functions*

- Simulated **6+ decision scenarios** under uncertainty to compute **Expected Value of Perfect Information (EVPI)** and **Expected Value of Sample Information (EVSI)**, quantifying the monetary ceiling on information acquisition.
- Applied utility-based and decision-theoretic frameworks to rank information sources by value, demonstrating a **2.4× cost-effectiveness improvement** in two illustrative trading contexts.

Technical Skills

Programming Languages: Python, R, C, C++, YAML

Quantitative & Statistical: Monte Carlo Simulation, Probability Theory, Statistical Modelling, Risk Assessment, Causal Inference, Decision Theory, Financial Mathematics, Algorithmic Trading

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Scikit-learn

Developer Tools: Git, GitHub, PyCharm, Jupyter Notebook

Finance Domain: Quantitative Analysis, Risk Modelling, Kelly Criterion, Probability Calibration, Backtesting, Value-at-Risk (VaR)

Certifications

Akuna Capital Options 101 — Akuna Capital University **2026**

Citi Markets Quantitative Analysis (MQA) — Forage **2025**

JPMorgan Quantitative Research Virtual Experience — Forage **2025**

Algorithmic Trading with Zerodha Kite Connect and Python — Quantra **2025**

Stock Market Basics — Zerodha Varsity **2025**

Achievements & Extra-Curricular Activities

Fincortex — IIT Roorkee (COMET'26 — CDC) **2026**

First Runner-Up | Finance & Technology Competition (Sponsored by Zaggle) *National Level*

- Secured **1st Runner-Up** among national participants in a premier fintech competition spanning algorithmic trading, fintech innovations,and quantitative problem-solving.

Techkriti — IIT Kanpur (TOSC Competition) **2022**

*All India Rank **18*** (Top 0.1% nationally) *National Level*

- Secured AIR 18 in the Test of Scientific Cognition (TOSC) at Techkriti, IIT Kanpur — a multi-stage national analytical competition testing quantitative reasoning and logical problem-solving under time constraints.

Science Olympiad Foundation (SOF) Olympiads **2022 – 2023**

International & National Ranks across 3 Olympiads *International Level*

- Achieved **International Rank 12** in ISSO, **International Rank 52** in NCO, and **International Rank 178** in NSO — placing in the global top percentile across three independent STEM disciplines.